

Aoxuan (Douglas) Li

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Education

PhD in Computer Science

Expected Dec 2028

IMDEA Software Institute | Advisor: Dario Fiore

MS in Computer Science

Sep 2017 – Jun 2019

University of California, Los Angeles (UCLA) | GPA: 3.81/4.0

Thesis: *On Explicit Depth Robust Graphs*

BS in Computing

2013 – 2017

Macao Polytechnic Institute

Research Experience

PhD Student

Dec 2024 – Present

IMDEA Software Institute

- Research succinct proof systems, polynomial commitments, and verifiable computation, with a focus on privacy-preserving and trustworthy AI.
- Contributed to the case study for refinement-based verification of cryptographic protocols with quantitative values; co-author of the ACM CCS 2026 paper (to appear).

Incoming — Frontier AI Security Residency

Oct 2026 – Dec 2026

ERA, Cambridge, UK

Researcher

Aug 2021 – Aug 2024

Macao Polytechnic University, Macao; University of Bologna, Italy

- Studied Internet Computer Protocol architectures through blockchain simulation, advised by Prof. Gabriele D'Angelo.
- Researched auditable confidentiality for decentralized finance, distributed ledger technologies, and smart contract security.

Cryptography Researcher (Intern)

Aug 2023 – Nov 2023

Nethermind

- Optimized STARK-based proof systems, reducing argument proof size and verification complexity.

Graduate Research Assistant

Jan 2018 – Jun 2019

UCLA, Cryptography Group

- Investigated explicit depth-robust graph constructions and cryptographic protocols under Prof. Rafail Ostrovsky.

Selected Publications

Refinement-based Verification of Cryptographic Protocol with Quantitative Values

I. Rakotonirina, J. Gomez-Martinez, **A. Li**, P. Moreno-Sanchez, and C. Schneidewind.
ACM Conference on Computer and Communications Security (CCS), 2026 (to appear).

An Auditable Confidentiality Protocol for Blockchain Transactions

A. Li, G. D'Angelo, S.-K. Tang, F. Fang, and B. Gong.
Blockchain: Research and Applications, 2025.

Implementation and Preliminary Evaluation of an Auditable Confidentiality Mechanism for DeFi

A. Li, S.-K. Tang, and G. D'Angelo.
IEEE International Conference on Distributed Computing Systems Workshops (ICDCSW), 2023.

Simulation of the Internet Computer Protocol: The Next Generation Multi-Blockchain Architecture

L. Serena, **A. Li**, M. Zichichi, G. D'Angelo, S. Ferretti, and S.-K. Tang.
IEEE/ACM Symposium on Distributed Simulation and Real Time Applications (DS-RT), 2022.

Industry & Engineering Experience

Cryptographer

Nov 2020 – Dec 2024

ExpandZK (Mystiko.Network); freelance from Aug 2021

- Designed the core protocol for Mystiko.Network; the platform recorded over \$280M in trading volume and 1M transactions during its first year.
- Designed a decentralized auditing committee mechanism and optimized ZK-rollup batching, reducing transaction fees by up to 90% compared with transactions without the protocol.

Software Engineer

Jul 2019 – Jul 2020

Stealth Software Technologies, Los Angeles

- Designed a private set intersection (PSI) algorithm and implemented cryptographic primitives in Java.
- Contributed to PULSAR/VLDS (Verifiable Lean Database Systems) and SAFRN (Secure Architecture for Resilient Networks).

Graduate Research Assistant

Jan 2018 – Jun 2019

UCLA, Cuneiform Digital Library Initiative

- Managed CentOS/MySQL infrastructure, optimized Docker backup workflows, and developed a CakePHP framework for digital heritage data.

Technical Skills

Programming: Python, Java, SQL.

Tools & Systems: Docker, Linux/CentOS, MySQL, $\text{\LaTeX}$.

Research Areas: Succinct proofs, polynomial commitments, verifiable computation, private set intersection, and distributed systems.

Academic Service

Subreviewer: Eurocrypt (2026, 2019), CHES (2026), INFOCOM (2023).

Teaching Assistant: Security and Cryptography; Optimization Methods, MPU.